CodeAlpha IoT Task 2

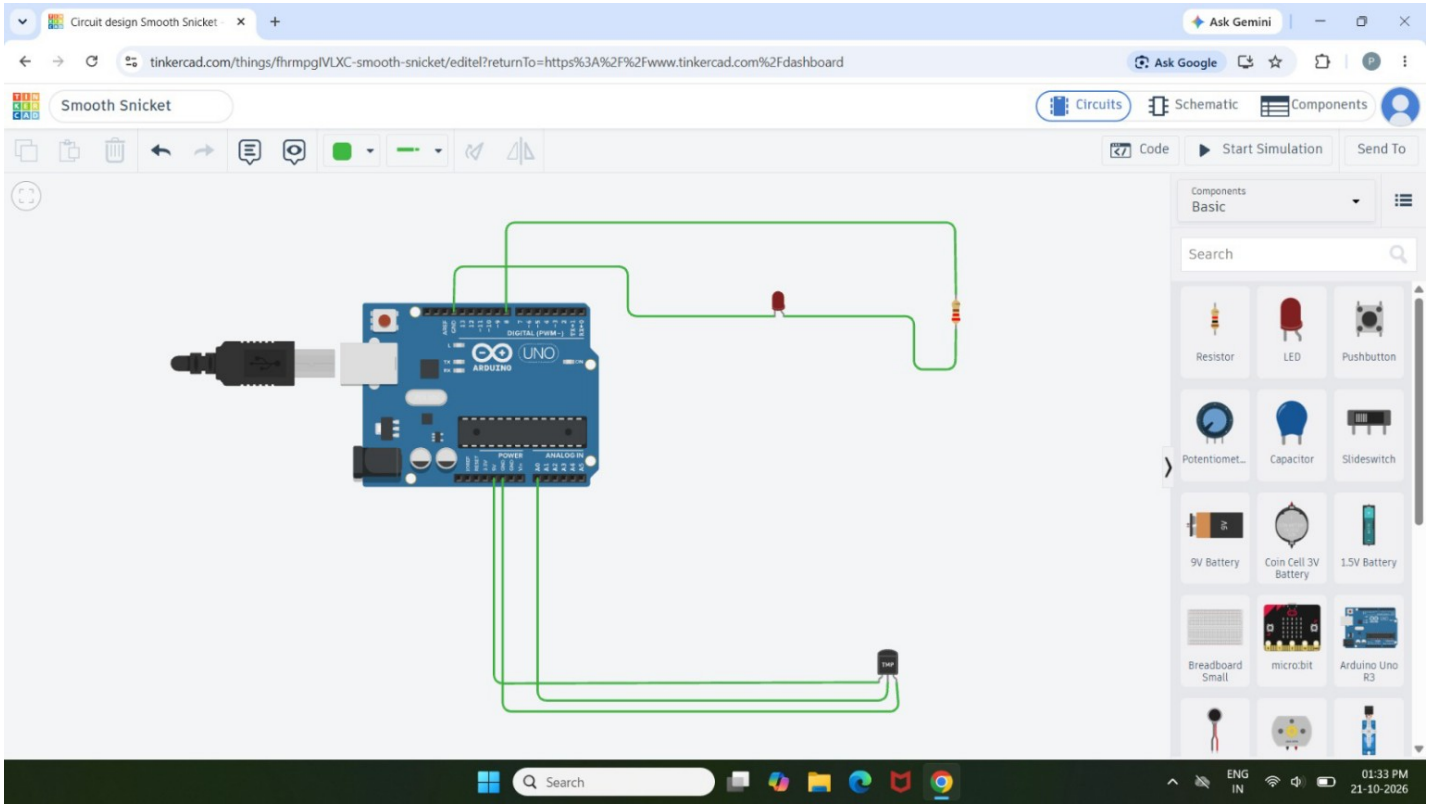
Sensor-Based Simulation: Temperature-Based LED Control Using Arduino and TMP36

Project Overview

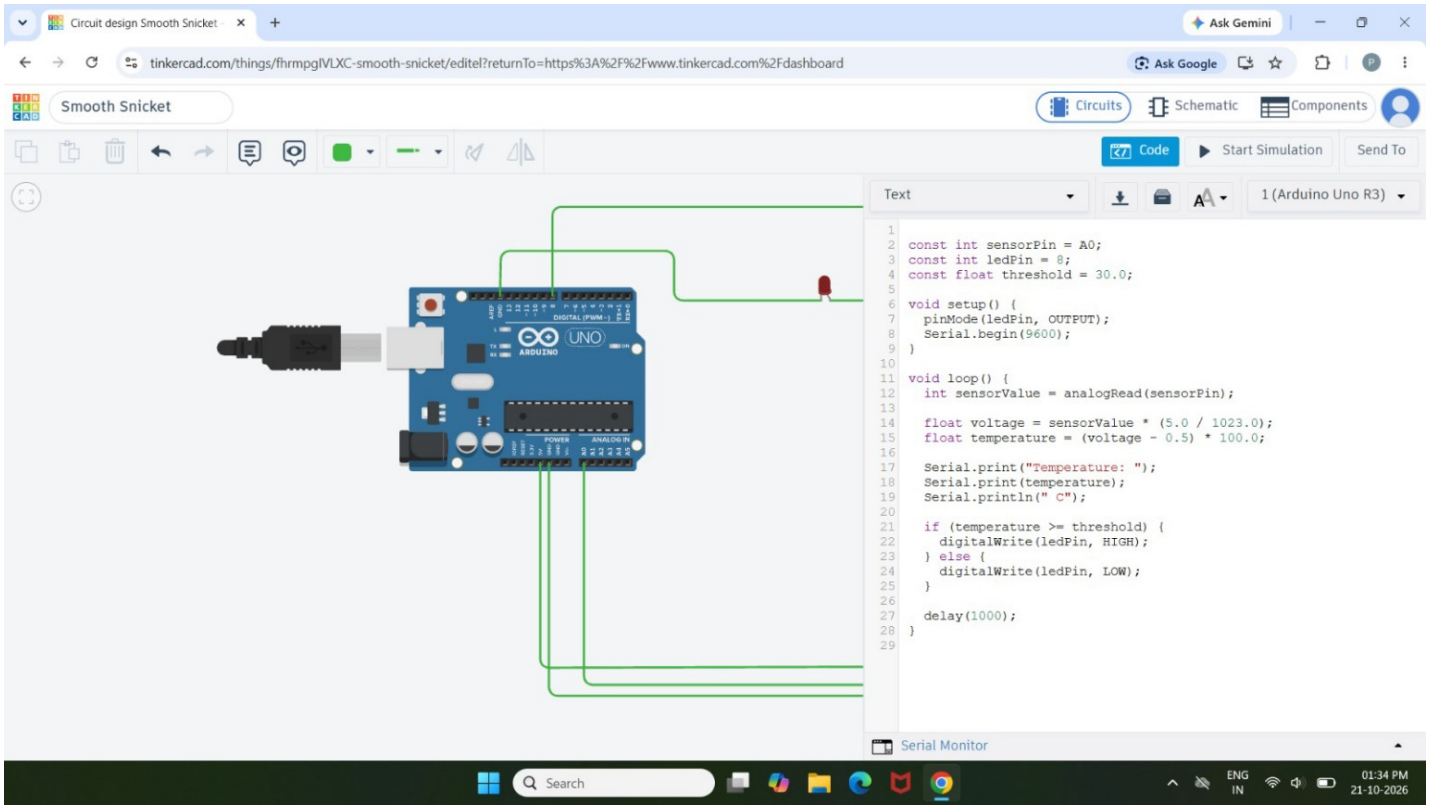
This simulation uses an Arduino Uno and a TMP36 temperature sensor to control an LED automatically. The Arduino reads the sensor output through analog pin A0, converts the reading to degrees Celsius, and compares it with a threshold of 30°C. When the temperature reaches or exceeds 30°C, the program turns the LED on; otherwise, it turns the LED off. Temperature readings are displayed in the Serial Monitor.

Simulation Screenshots

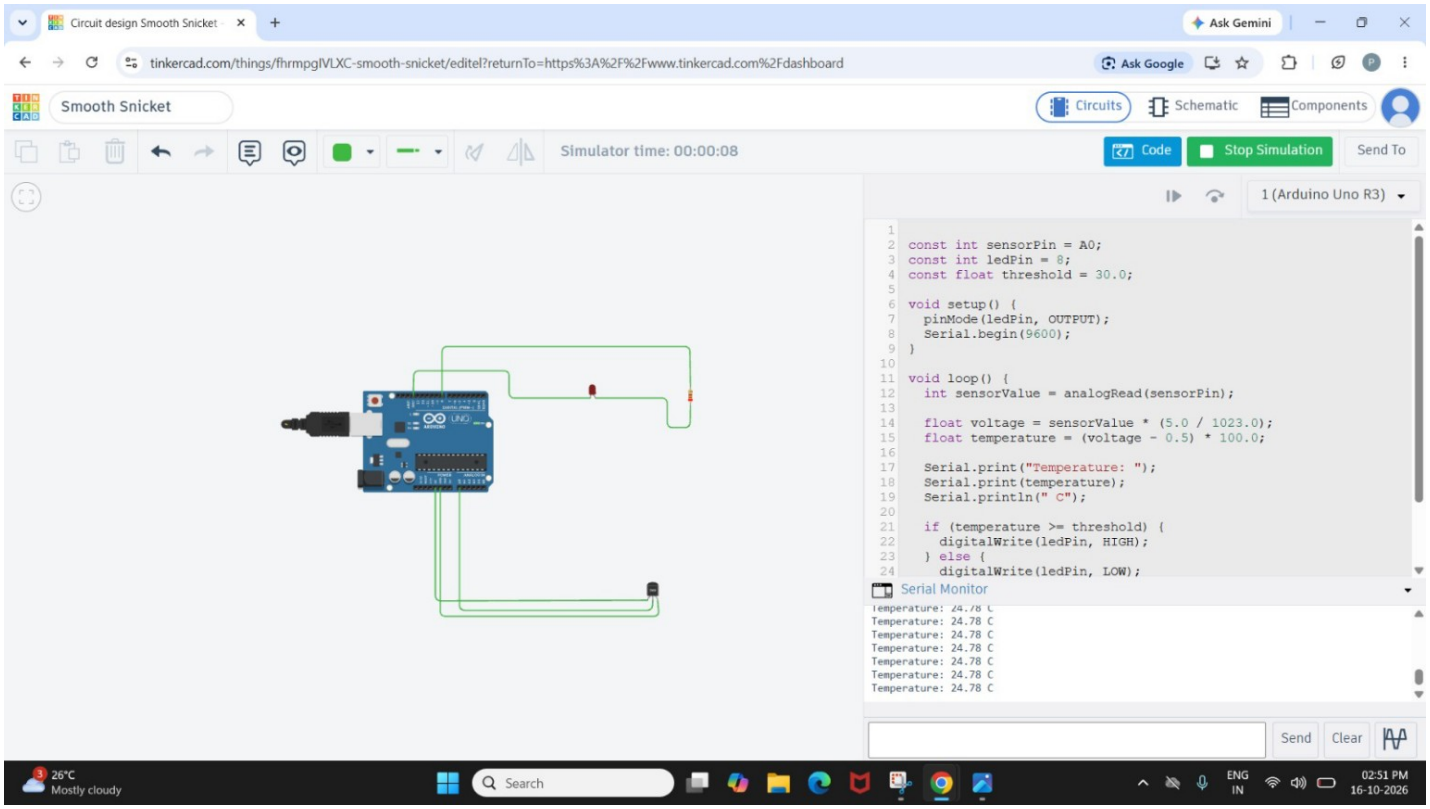
Screenshot 1: Tinkercad circuit layout.



Screenshot 2: Arduino code displayed beside the circuit.



Screenshot 3: Running simulation and Serial Monitor temperature readings.



Screenshot 4: Temperature raised above the 30°C threshold; LED turns on.

